Unit 5, Lesson 8: Triangles on the Coordinate Plane

Benchmark

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Learning Targets

- ☐ **I can determine the horizontal and vertical lengths of a triangle on a coordinate plane.**
- ☐ **I can determine the hypotenuse length of a triangle on the coordinate plane.**

5.8.1 Warm-Up: Critical Thinking

1. Use any combination of operations (+, −, x, ÷) to arrange five 5's to have an equivalent value of 0, 1, 2, 3, 4, and 5. Multiple 5's can be used in one number (e.g. 55).

   Example: $(5 \times 5 - 5 \times 5) \times 5 = 0$

   0 =

   3 =

   1 =

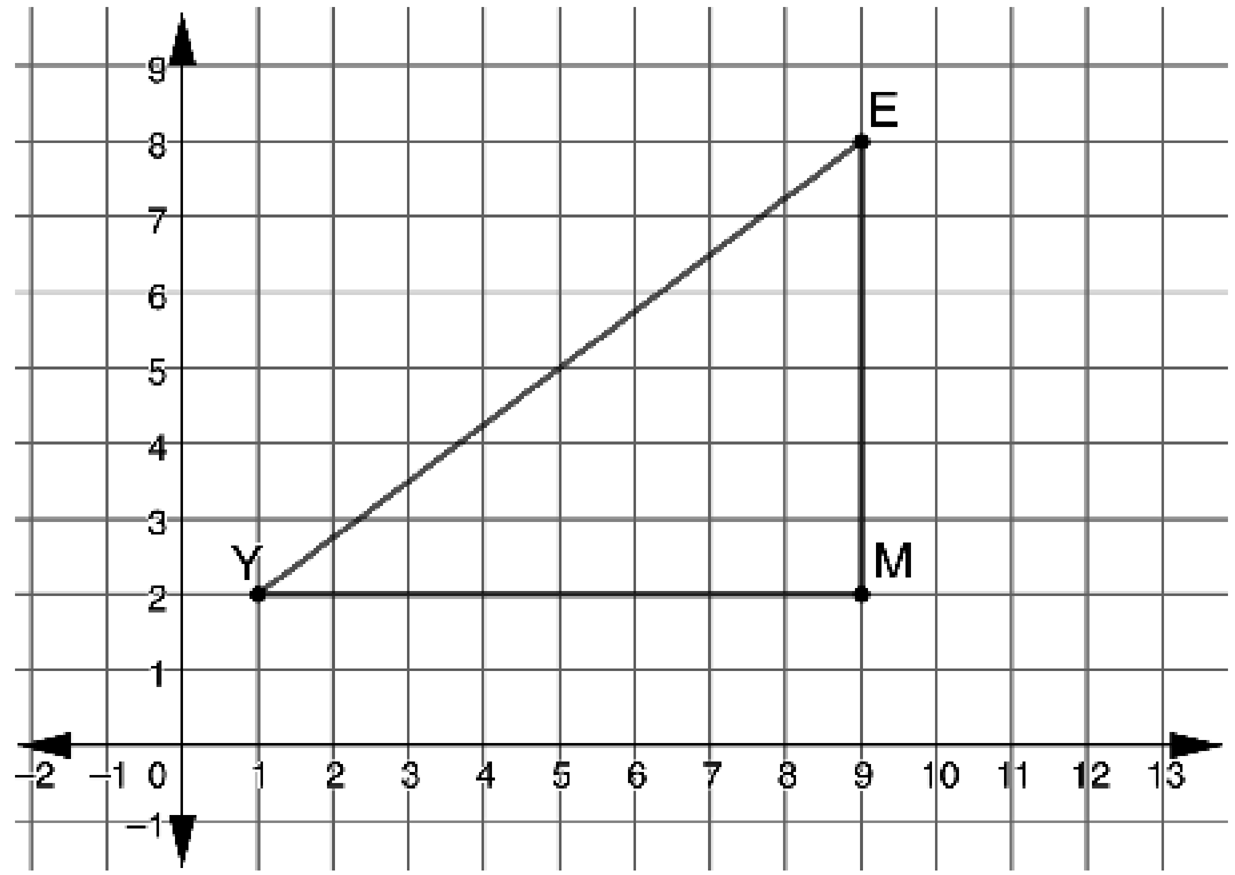
   4 =

   2 =

   5 =

5.8.2 Guided Instruction: Finding the Length

The Triangle *EMY* is shown on the coordinate plane.



1. List the ordered pairs for each vertex.

   a. Point *E* (____, ____)

   b. Point *M* (____, ____)

   c. Point *Y* (____, ____)

2. What is the length of the horizontal leg of Triangle *EMY*?

3. What do you notice about the horizontal length and the coordinates of point *M* and point *Y*?

4. What is length of the vertical leg of Triangle *EMY*?

5. What do you notice about the vertical length and the coordinates of point *E* and point *M*?

6. Complete the statements.

To determine horizontal length on the coordinate plane,

find the difference between the
- o x-coordinates.
- o y-coordinates.

To determine vertical length on the coordinate plane,

find the difference between the
- o x-coordinates.
- o y-coordinates.

5.8.3 Try It: Finding Horizontal and Vertical Distances

1. Find the **horizontal** distance between the following points.

a. $(2, 7)$ and $(8, 7)$ __________ unit(s)

b. $(-5, 1)$ and $(3, 1)$ __________ unit(s)

c. $(0, 4)$ and $(6, -3)$ __________ unit(s)

2. Find the **vertical** distance between the following points.

a. $(2, -9)$ and $(2, 6)$ __________ unit(s)

b. $(-4, -1)$ and $(-4, -3)$ __________ unit(s)

c. $(0, 4)$ and $(6, -3)$ __________ unit(s)

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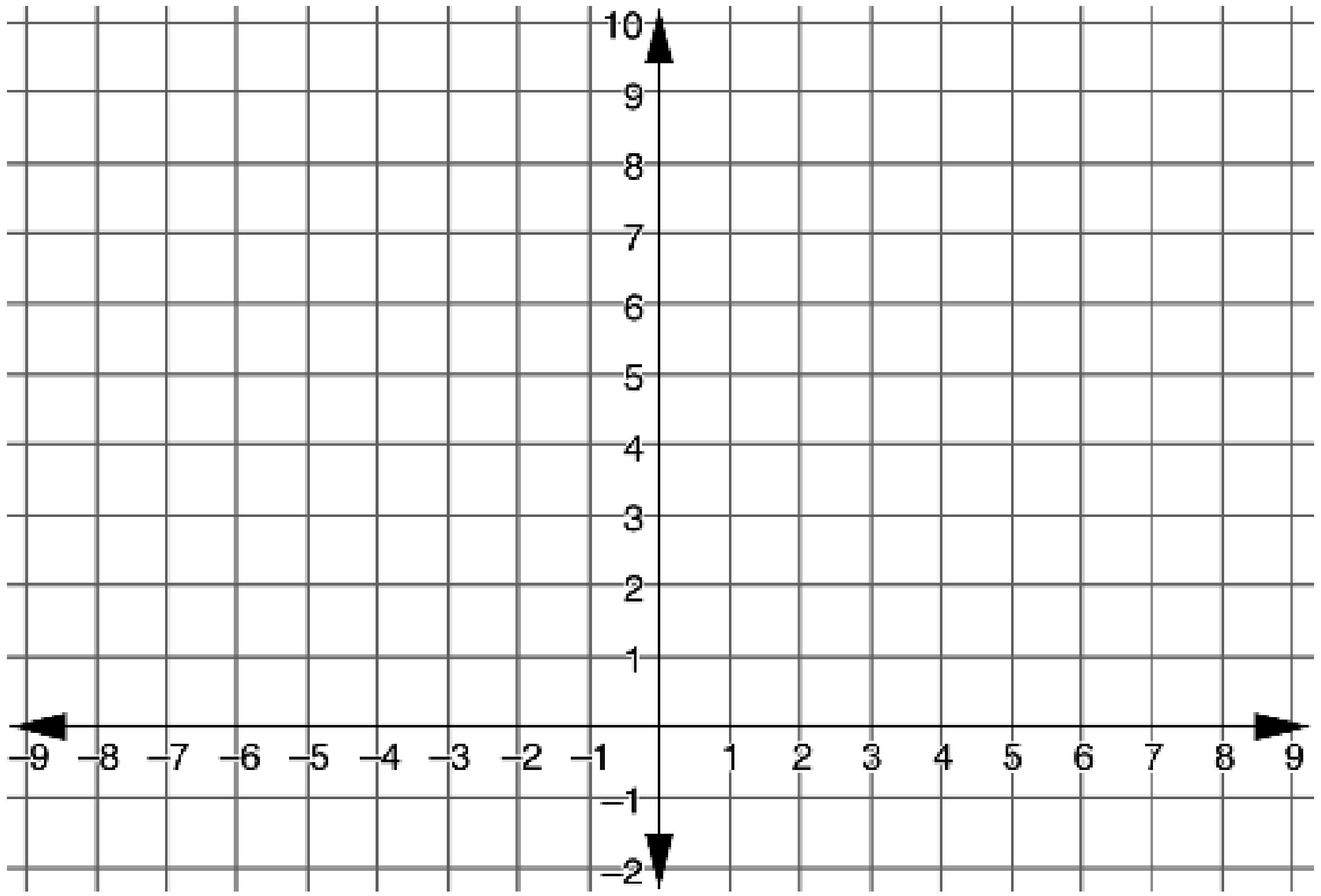
5.8.4 Guided Instruction: Right Triangles on the Coordinate Plane

1. A triangle is defined by the following vertices in the coordinate plane.

Point B (8, 3) Point H (−2, 3) Point L (8, 7)

a. Graph Triangle BHL.



b. Find the length of the horizontal leg and label the leg with its length.

c. Find the length of the vertical leg and label the leg with its length.

d. Using the Pythagorean Theorem, find the length of the hypotenuse ($\overline{HL}$). Round to the nearest tenth, if necessary.

e. Find the perimeter of the triangle.

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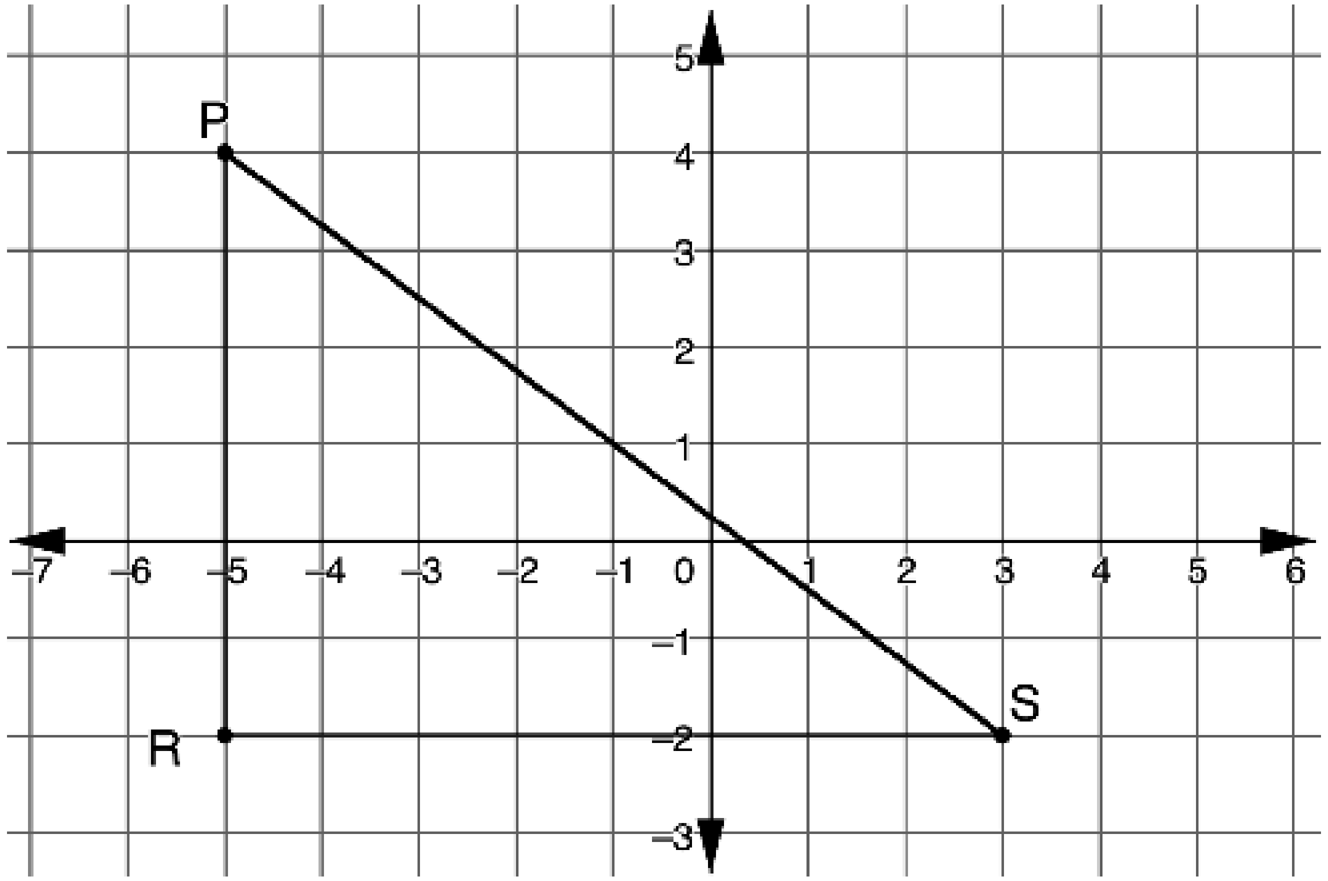
5.8.5 Try It: Right Triangles on the Coordinate Plane

Triangle *PRS* is graphed on the coordinate plane.



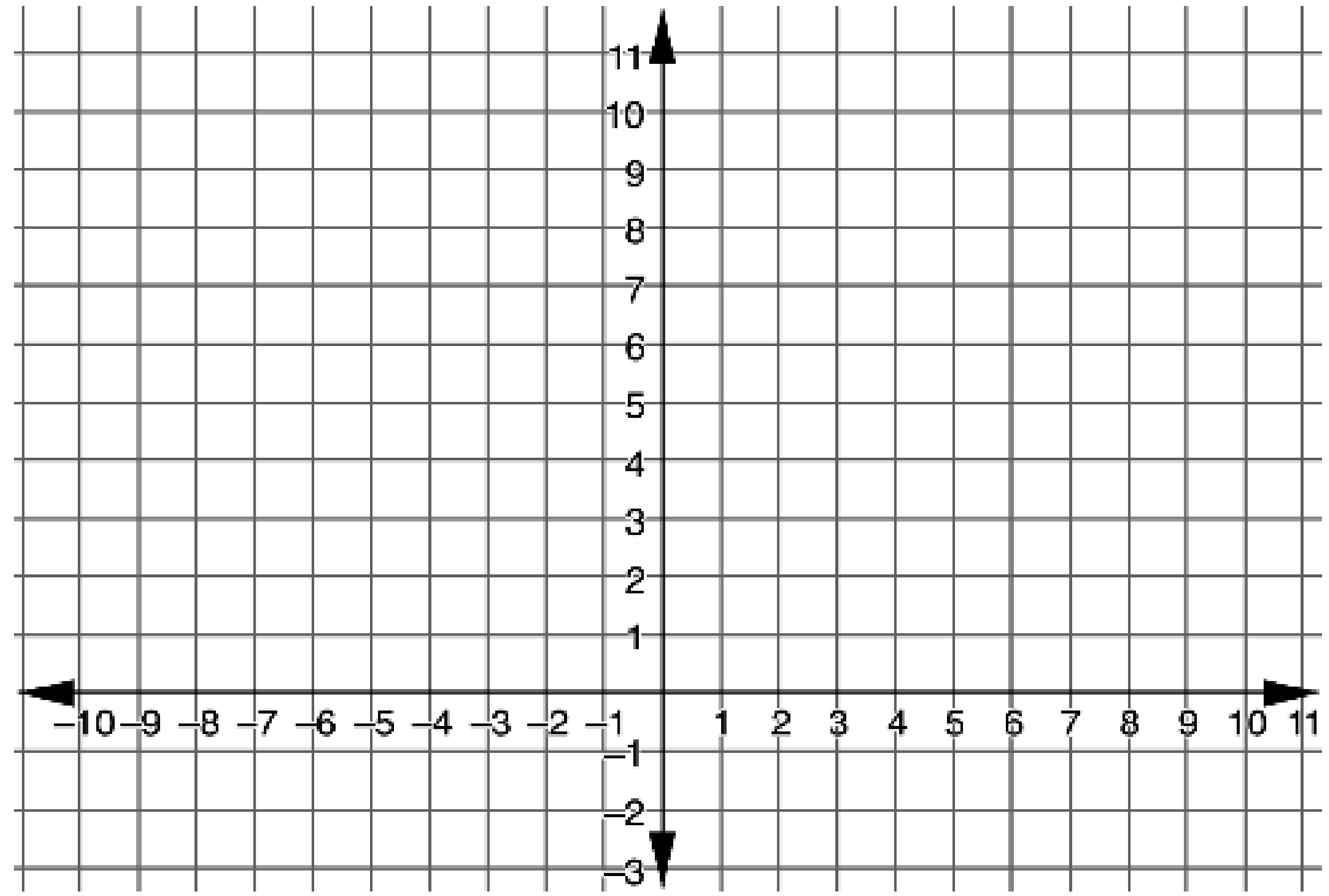
1. Find the perimeter of Triangle *PRS*. Round to the nearest tenth, if necessary.

5.8.6 Guided Practice: Finding Perimeter

1. A triangle is defined by the following vertices on the coordinate plane.

Point *W* (–4, 2) Point *P* (8, 2) Point *S* (2, 10)

a. Graph Triangle *WPS*.



b. What type of triangle is Triangle *WPS*?

c. Find the horizontal length of Triangle *WPS*.

d. Draw a perpendicular line segment from point S to $\overline{WP}$.

e. Use the Pythagorean Theorem to find the lengths of $\overline{WS}$ and $\overline{PS}$.

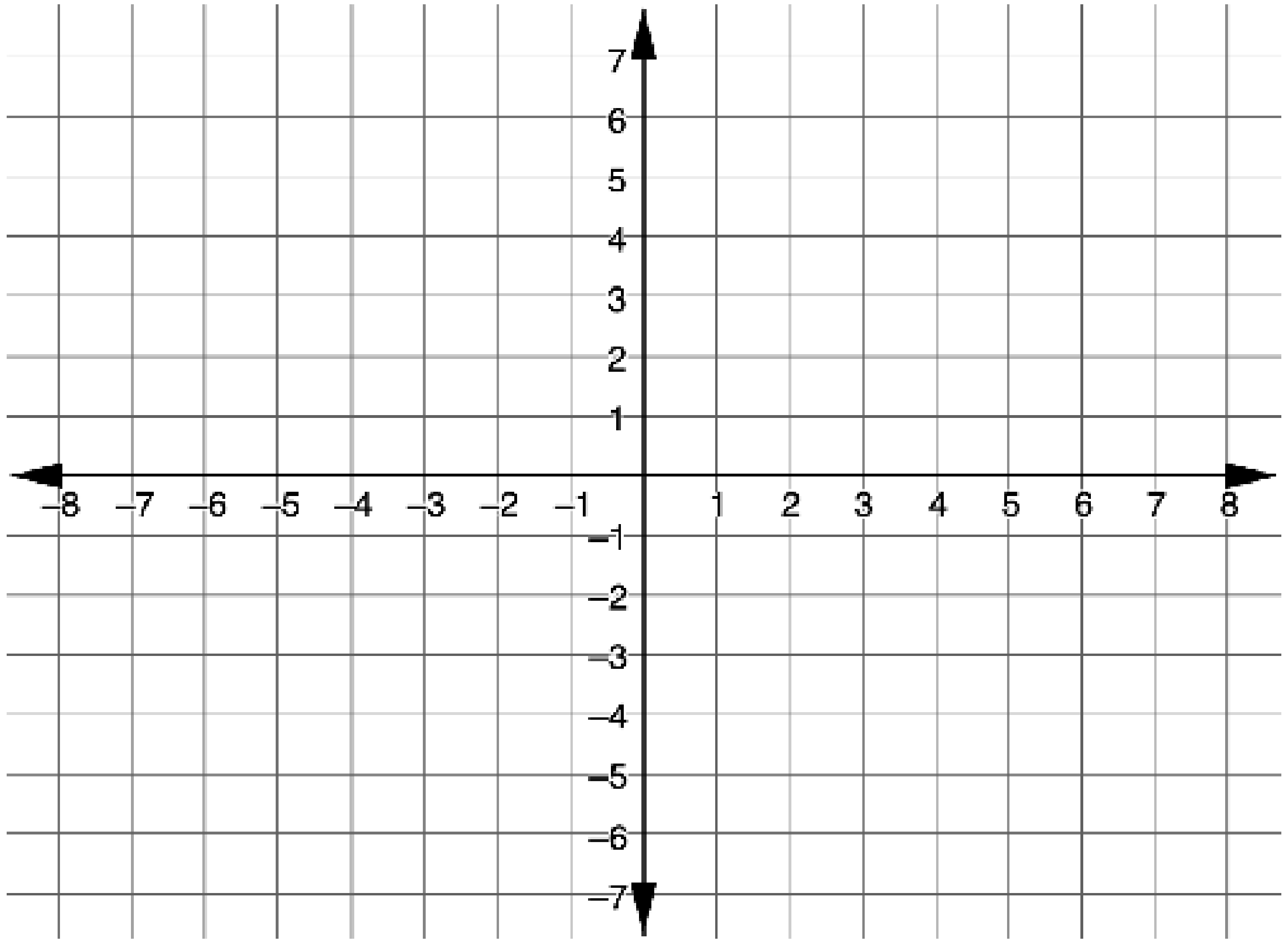
f. Find the perimeter of Triangle WPS.

5.8.7 Your Turn!

1. Triangle DKF is defined by the following vertices.

$D\,(-1,0)$ $K\,(2,-6)$ $F\,(2,0)$

a. Graph Triangle DKF.



b. Label each side with its length. Round to the nearest tenth, if needed.

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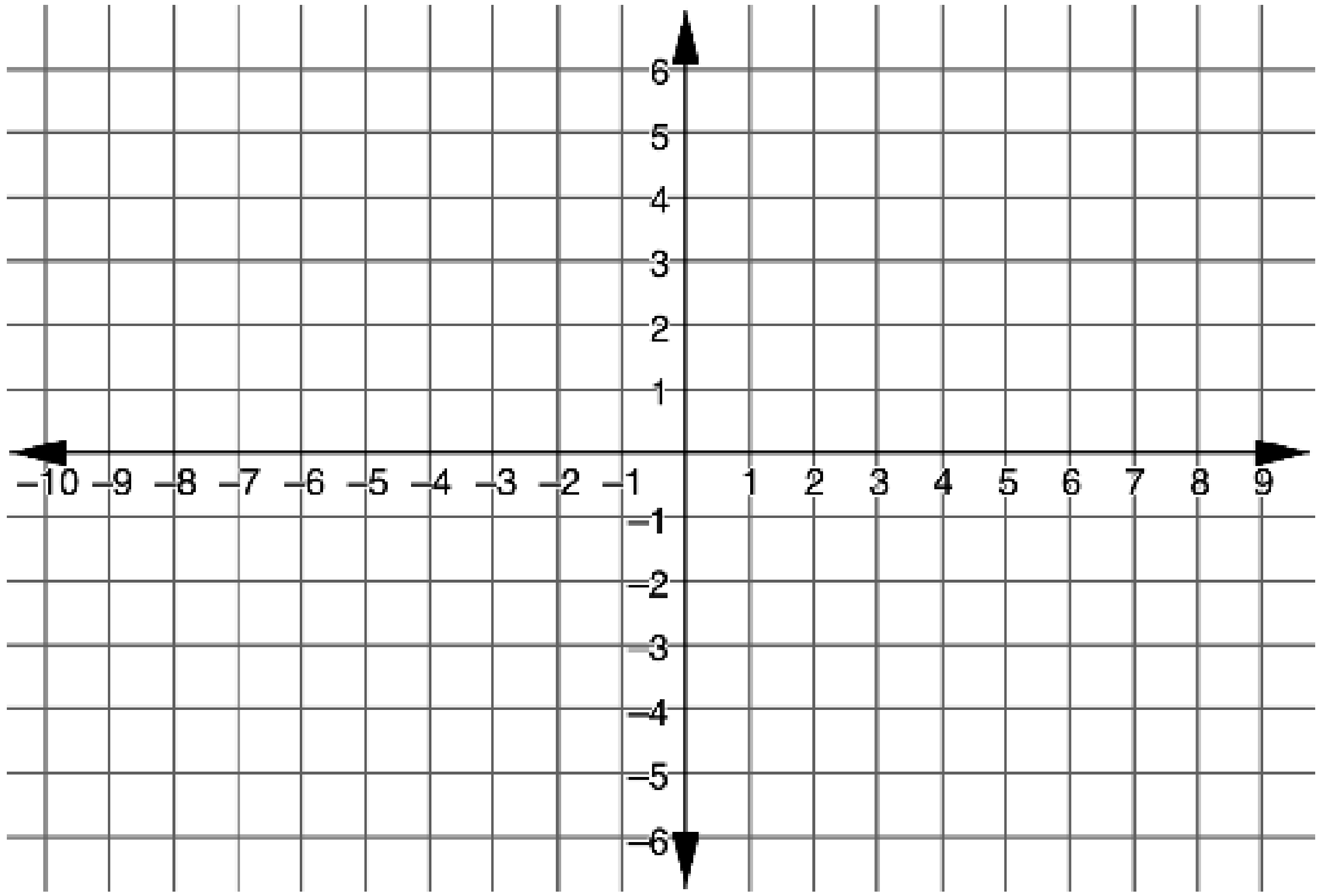
2. Triangle UHG is defined by the vertices.

$U\ (-9, 3)$ $H\ (6, 3)$ $G\ (3, -5)$

Find the perimeter of Triangle UHG. Round to the nearest tenth, if needed.



5.8.8 Wrap-Up: Reflection

1. Place an X on each line to indicate your understanding of each success criteria.

Success Criteria	Self-Assessment		
I can apply the Pythagorean Theorem to find the distance of a segment on the coordinate plane.	I need help. I can't get started.	I'm getting there.	I understand and I have evidence.
I can find the perimeter of a right triangle on the coordinate plane.	I need help. I can't get started.	I'm getting there.	I understand and I have evidence.